Basic Syntax and Common Errors in Nand2Tetris

【Basic Syntax】

1. File Structure

Each .hdl file describes a single chip.

Basic format:

```
CHIP ChipName {
    IN in1, in2, ...;
    OUT out1, out2, ...;

    PARTS:
    // Components go here
}
```

2. Input/Output Declarations

IN declares input pins.

OUT declares output pins.

You can define single bits or buses (bit arrays).

Examples:

```
IN a, b;              // two 1-bit inputs
OUT out;              // one 1-bit output
IN x[16], y[16];    // two 16-bit buses
OUT sum[16];          // one 16-bit bus
```

3. Using Built-in or Custom Chips

Under PARTS:, you instantiate other chips.

Example:

```
CHIP And16 {
    IN a[16], b[16];
    OUT out[16];

    PARTS:
    And(a=a[0], b=b[0], out=out[0]);
    And(a=a[1], b=b[1], out=out[1]);
    // ... etc.
}
```

4. Constants and Pins

You can use:

true or false

Specific bits of a bus, e.g. a[3]

Ranges of bits, e.g. a[0..7]

Example:

```
Not(in=false, out=alwaysTrue);          // out = true
And(a=a[0], b=true, out=out[0]);        // copy bit
Or(a=a[1], b=b[1], out=out[1]);         // OR operation
```

5. Bus Connections

Connect slices of buses or whole buses:

```
Mux(a=a[0], b=a[1], sel=sel, out=out);
Add16(a=x[0..15], b=y[0..15], out=out[0..15]);
```

6. A chip's output can be connected to multiple inputs of other chips.

```
CHIP Example {
    IN a, b;
    OUT out1, out2;

    PARTS:
    And(a=a, b=b, out=andOut);
    Not(in=andOut, out=out1);
    Or(a=andOut, b=b, out=out2);
}
```

【Common Errors】

1. Syntax & Formatting Errors

Mistake	Example	Fix
Missing semicolons	IN a, b	IN a, b;
Using Chinese punctuation marks (，；) instead of English punctuation marks (,) and (;).	IN a, b；	IN a, b;
Forgetting commas between pins	IN a b;	IN a, b;
Mismatched braces	CHIP MyChip { …	Must have closing }
Wrong keyword case	chip, in, out	Must be uppercase: CHIP, IN, OUT, PARTS
Missing PARTS:	parts list directly under OUT	Always include PARTS: line
Insert a space between PART and colon	PARTS :	PARTS:
Extra commas	IN a, b,;	Remove trailing comma
Using underline '_'	Inc16(in = feed_back, out = pc);	Inc16(in = feedback, out = pc);

Tip: HDL syntax is *case-sensitive* and *semicolon-sensitive*. The simulator will not auto-correct.

2. Connection & Naming Errors

Mistake	Example	Fix
Pin names don't match the chip's interface	And(a=x, b=y, output=z);	Use correct pin name: And(a=x, b=y, out=z);
Using undefined internal wires	Not(in=tmp, out=out); but tmp never defined	Use only declared pins or internal chip outputs
Reusing output pins as inputs accidentally	Output out used in both directions	Create intermediate wires if needed

Tip: Every connection must reference either an input pin, output pin, or another chip's output. Nothing "auto-wires."

3. Bus and Bit Indexing Errors

Mistake	Example	Fix
Wrong index syntax	a[0-7]	Must use a[0..7]
Index out of range	a[16] for a[16] bus	Highest valid index is 15
Mismatched bus widths	And16(a=a[16], b=b, out=out);	All bus widths must match

Mixing single bits with buses	And(a=a, b=b[0..15], out=out);	Use single-bit or full-bus chips consistently

Tip: HDL doesn't auto-expand or shrink buses; you must connect the right widths explicitly.

4. Logic or Behavioral Mistakes

Mistake	Example	Why it's wrong
Swapped input/output pins	Not(out=in, in=out);	Outputs are write-only; order matters
Forgetting to connect all required pins	Mux(a=a, b=b, out=out); (missing sel)	All pins must be explicitly connected
Connecting multiple sources to same wire	Two chips write to the same output pin	Illegal — only one chip can drive a wire
Misusing constants	a=1 or a=0	Must use true or false

5. Simulation & Testing Pitfalls

Mistake	Example	Fix
Using wrong chip name in test file	File And.hdl but CHIP AND inside	Chip name in file must match the .hdl filename exactly
Editing built-in chips	Modifying And.hdl instead of creating your own	Only edit project chips; built-ins are for reference
Not saving before loading in simulator	HDL not reloaded	Always save before reloading or testing
Ignoring "Comparison failed" output	Assuming the tool is wrong	The simulator is always right — check bit order, naming, or bus size